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Sec: B, Roll No. 53 LABSHEET 03

```python
import pandas as pd

df = pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-selected-
columns.csv")

print(df.head())
print(df.shape)
print(df.columns.tolist())
```

```
   Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7
(1470, 10)
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount', 'EmployeeNumber']
```

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
import statsmodels.api as sm
```

```python
print("Mean:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].mean())

print("\nMedian:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].median())

print("\nMode:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].mode().iloc[0])

print("\nVariance:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].var())

print("\nStandard Deviation:")
print(df[["Age", "DailyRate", "DistanceFromHome"]].std())
```

```
Mean:
Age                   36.923810
DailyRate            802.485714
DistanceFromHome       9.192517
dtype: float64

Median:
Age                   36.0
DailyRate            802.0
DistanceFromHome       7.0
dtype: float64

Mode:
Age                   35
DailyRate            691
DistanceFromHome       2
Name: 0, dtype: int64

Variance:
Age                       83.455049
DailyRate             162819.593737
DistanceFromHome          65.721251
dtype: float64

Standard Deviation:
Age                    9.135373
DailyRate            402.509100
DistanceFromHome       8.106864
dtype: float64
```

```python
print(df[["Age", "DailyRate", "DistanceFromHome"]].describe())
```

```
count              Age    DailyRate  DistanceFromHome
mean       1470.000000  1470.000000       1470.000000
std          36.923810   802.485714          9.192517
              9.135373   403.509100          8.106864
```

min	18.000000	102.000000	1.000000
25%	30.000000	465.000000	2.000000
50%	36.000000	802.000000	7.000000
75%	43.000000	1157.000000	14.000000
max	60.000000	1499.000000	29.000000

```
numeric_df = df[["Age", "DailyRate", "DistanceFromHome"]]
correlation = numeric_df.corr()

print(correlation)
```

Age

Age DailyRate		DistanceFromHome
DailyRate 0.010661	1.000000	-0.001686
1.000000	0.010661	-0.004985
DistanceFromHome -0.001686 -0.004985		1.000000

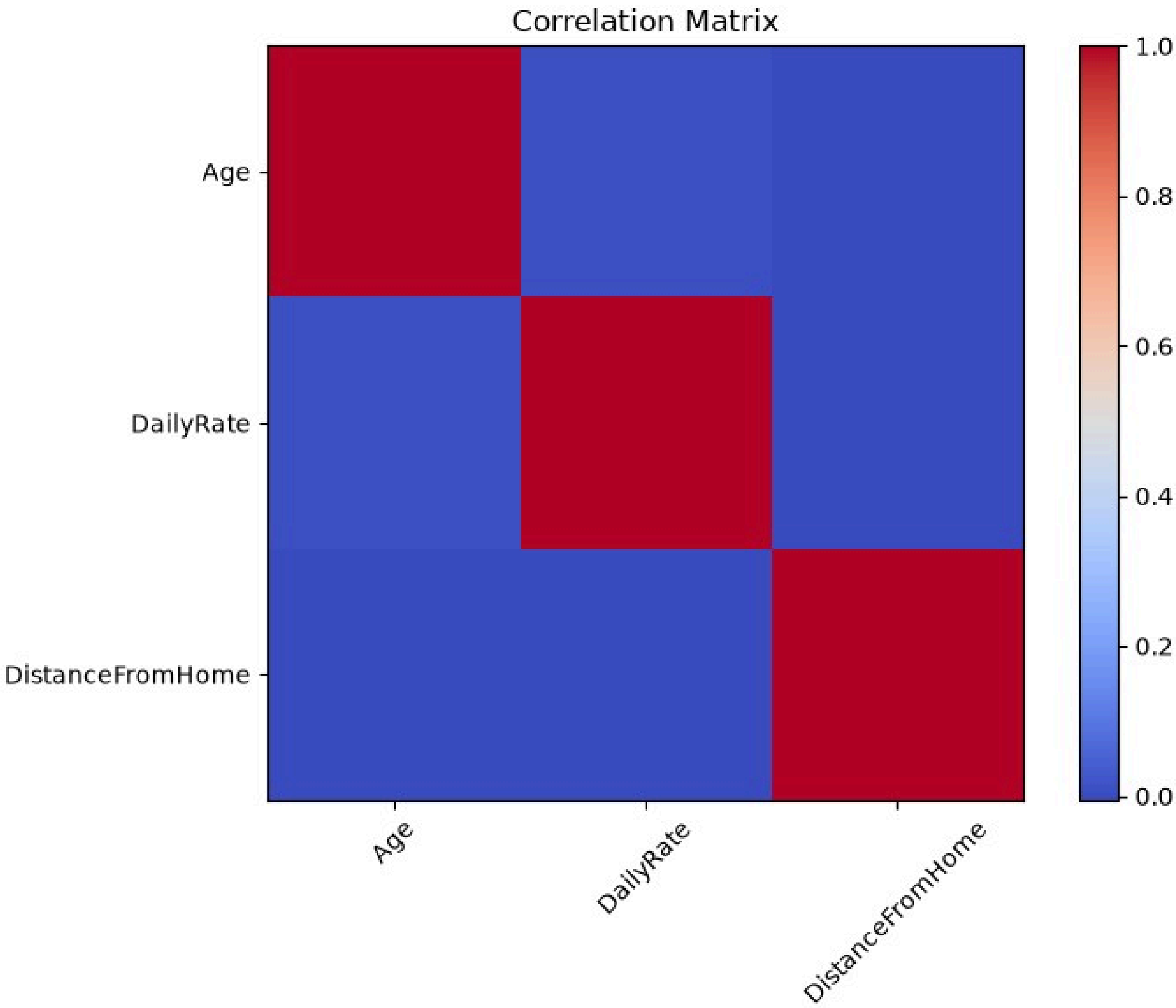
```
plt.figure(figsize=(8, 6))

plt.imshow(correlation, cmap="coolwarm")
plt.colorbar()

plt.xticks(
        range(len(correlation.columns)),
        correlation.columns,
        rotation=45
)

plt.yticks(
        range(len(correlation.columns)),
        correlation.columns
)

plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```




```
r, p = stats.pearsonr(df["Age"], df["DailyRate"])

print("Pearson Correlation:", r)
print("P-value:", p)

Pearson Correlation: 0.01066094264553816
P-value: 0.6829722453880914

left = df[df["Attrition"] == "Yes"]["Age"]

stayed = df[df["Attrition"] == "No"]["Age"]
print("Employees who left:", len(left))
print("Employees who stayed:", len(stayed))
Employees who left: 237
Employees who stayed: 1233
```

```
t_stat, p_value = stats.ttest_ind(
        left,
        stayed,
        equal_var=False
)

print("T-statistic:",    t_stat)    print("P-
value:", p_value)
```

```
T-statistic:    -5.82801185398895    P-
value: 1.3797600649439802e-08
```

```
if p_value < 0.05:
        print("Reject H0")
        print("There is a significant difference in Age between employees
who left and stayed.")
else:
        print("Fail to reject H0")
        print("There is no significant difference in Age between employees
who left and stayed.")
```

```
Reject H0
There is a significant difference in Age between employees who left
and stayed.
```

```
print(df["Department"].unique())
```

```
<StringArray>
['Sales', 'Research & Development', 'Human Resources']
Length: 3, dtype: str
```

```
groups = []

for department in df["Department"].unique():
        groups.append(
                df[df["Department"] == department]["Age"]
        )
```

```
f_stat, p_value_anova = stats.f_oneway(*groups)

print("F-statistic:", f_stat)
print("P-value:", p_value_anova)
```

```
F-statistic: 0.7655031924976907
P-value: 0.4652855299935297
```

```
if p_value_anova < 0.05:
        print("Reject H0")
        print("There is a significant difference in Age among
Departments.")
else:
        print("Fail to reject H0")
```

```
    print("There is no significant difference in Age among Departments.")
```

```
Fail to reject H0
There is no significant difference in Age among Departments.
```

```
X = df["Age"]
y = df["DailyRate"]

X = sm.add_constant(X)

model = sm.OLS(y, X).fit()

print(model.summary())
```

```
                    OLS Regression Results
==============================================================================
Dep. Variable:                DailyRate   R-squared:
0.000
Model:                              OLS   Adj. R-squared:   F-
-0.001
Method:                   Least Squares   statistic:
0.1669
Date:              Mon,31 Aug 2026   Prob (F-statistic): Log-
0.683
Time:                          08:31:36   Likelihood: AIC:
-10906.
No. Observations:                  1470   BIC:
2.182e+04
Df Residuals:                      1468
2.183e+04
Df Model:                             1

CovarianceType:               nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
const        785.0985     43.847     17.905      0.000     699.089
871.108
Age            0.4709      1.153      0.408      0.683      -1.790
2.732
==============================================================================
Omnibus:                   1174.453   Durbin-Watson:
2.082
```

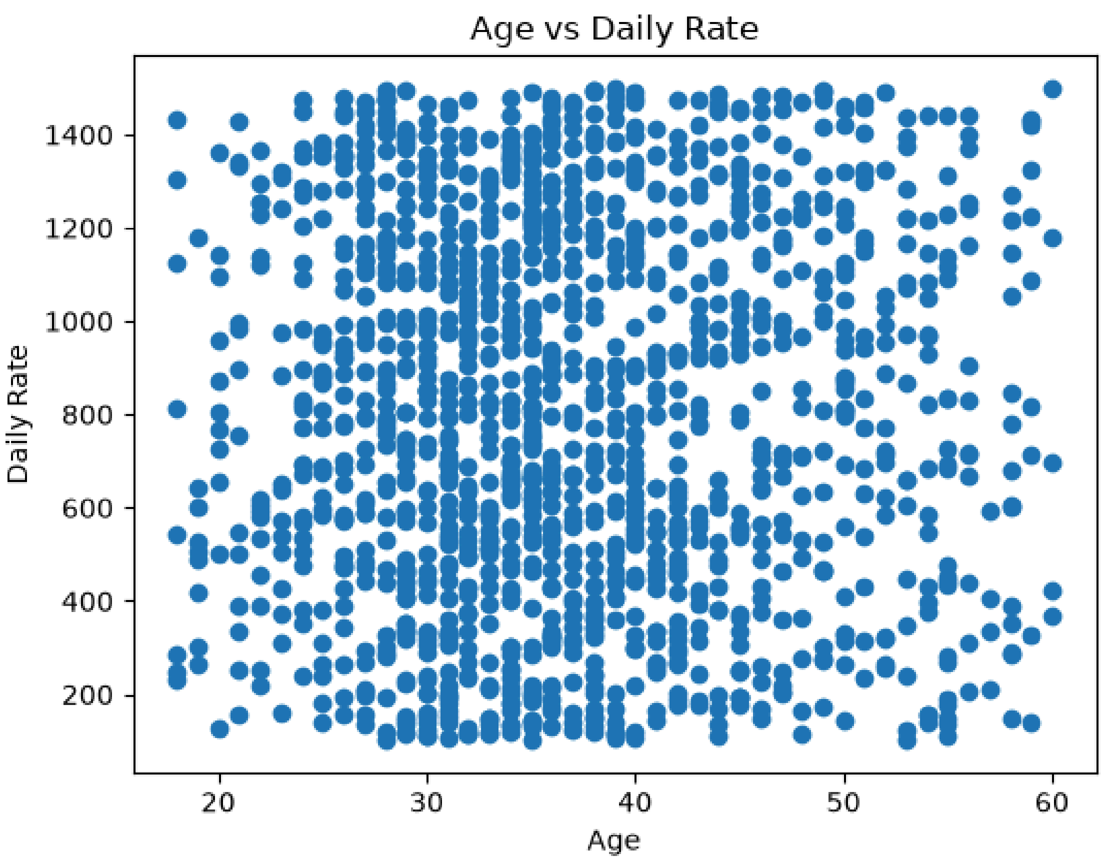
```
Prob(Omnibus):                     0.000   Jarque-Bera   (JB):
88.696
Skew:                              -0.004   Prob(JB):
5.49e-20
Kurtosis:                          1.797   Cond. No.
159.
==============================================================
=================== ========

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.

plt.scatter(df["Age"], df["DailyRate"])

plt.xlabel("Age")
plt.ylabel("Daily Rate")
plt.title("Age vs Daily Rate")

plt.show()
```




```
print("R-squared:", model.rsquared)
```

```
R-squared: 0.00011365569809163034
```

```python
print("Regression Coefficients:")
print(model.params)
```

```
Regression Coefficients:
const    785.098535
Age        0.470893
dtype: float64
```

```python
print("P-values:")
print(model.pvalues)
```

```
P-values:
const    5.267571e-65
Age      6.829722e-01
dtype: float64
```

```python
print("Confidence Intervals:")
print(model.conf_int())
```

```
Confidence Intervals:
                0           1
const   699.089362  871.107708
Age      -1.790341    2.732128
```

```python
print(model.summary())
```

```
                            OLS Regression Results
============================================================
========
Dep. Variable:                 DailyRate   R-squared:
0.000
Model:                               OLS   Adj. R-squared:   F-
-0.001
Method:                    Least Squares   statistic:
0.1669
Date:                  Mon,31 Aug 2026   Prob (F-statistic): Log-
0.683
Time:                           08:31:37   Likelihood: AIC:
-10906.
No. Observations:                   1470   BIC:
2.182e+04
Df Residuals:                       1468
2.183e+04
Df Model:                              1

CovarianceType:                nonrobust

============================================================
```

```
========
                      coef      std err            t        P>|t|          [0.025
0.975] ------------------------------------------------------------------ ---------
const
871.108
Age                785.0985       43.847        17.905       0.000        699.089
2.732
======================0.4709======1.153=======0.408=======0.683======-1.790========
=================== ========
Omnibus:
2.082
Prob(Omnibus):                        1174.453    Durbin-Watson:
88.696
Skew:                                    0.000    Jarque-Bera    (JB):
5.49e-20
Kurtosis:                               -0.004    Prob(JB):
159.
================================1.797===Cond. No.=====================
=================== ========

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

```python
print("STATISTICAL ANALYSIS SUMMARY")
print("=" * 40)

print("\n1. T-TEST")
print("T-statistic:", t_stat)
print("P-value:", p_value)

if p_value < 0.05:


    print("Result: Significant difference in Age.")
else:
    print("Result: No significant difference in Age.")

print("\n2. ANOVA")
print("F-statistic:", f_stat)
print("P-value:", p_value_anova)

if p_value_anova < 0.05:
    print("Result: Significant difference among Departments.")
else:
    print("Result: No significant difference among Departments.")

print("\n3. LINEAR REGRESSION")
print("R-squared:", model.rsquared)
```

```
print("Age coefficient:", model.params["Age"])
print("Age p-value:", model.pvalues["Age"])
```

```
STATISTICAL ANALYSIS SUMMARY
==================================================

1. T-TEST
T-statistic: -5.82801185398895
P-value: 1.3797600649439802e-08
Result: Significant difference in Age.

2. ANOVA
F-statistic: 0.7655031924976907
P-value: 0.4652855299935297
Result: No significant difference among Departments.

3. LINEAR REGRESSION
R-squared: 0.00011365569809163034
Age coefficient: 0.47089343162226616
Age p-value: 0.6829722453880929
```